Introduction to FFLO Superconductivity

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Outline

BCS

FFLO

Experiments

Pairing symmetry

$$V_{\text{eff}}(r) < 0$$

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$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_1, \mathbf{r}_2) \otimes \chi(s_1, s_2)$$

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$$\chi_s(s_1, s_2) = -\chi_s(s_2, s_1)$$

$$\rightarrow \psi_s(1,2) = \psi_s(2,1)$$

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$$\Psi(1,2) = -\Psi(2,1)$$

$$\langle V \rangle_t > \langle V \rangle_s$$

Cooper Pair

$$\mathcal{H} = \sum_{k,\sigma} \epsilon_k \hat{n}_{k,\sigma} + \sum_{k,k'} V_{k,k'} \hat{c}^\dagger_{k\uparrow} \hat{c}^\dagger_{-k\downarrow} \hat{c}_{k'\uparrow} \hat{c}_{-k'\downarrow}$$

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$$V^{-1} = \sum_{|k'|>k_F} (E - 2\epsilon_k)^{-1}$$

Ground State

hello

Band Gap

hello

Zeeman Splitting

hello

Fermi shell

hello

New Auxiliary Field

hello

Hard To Measure

hello

EXP 1

hello

EXP 2

hello